Lecture 7

1. Transformation & Terminology

Def: A transformation T (functions, mappings) from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns to each vector $\vec{x} \in \mathbb{R}^n$ a vector $T(\vec{x}) \in \mathbb{R}^m$

- $\vec{x} \in \mathbb{R}^n$ (input) $\longrightarrow T(\vec{x}) \in \mathbb{R}^m$ (output)
- $T: \mathbb{R}^n \to \mathbb{R}^m$ (says that T is a **function** from $\mathbb{R}^n \to \mathbb{R}^m$)
- $x \mapsto y$ (x maps to y, $T(\vec{x}) = \vec{y}$)

$\mathbb{R}^n$: **domain** of T

$\mathbb{R}^m$: **Codomain** of T

$T(\vec{x})$: "image of $\vec{x}$ under T"

Def: The set of all things that T outputs is called **the range** of T

$$\text{Range}(T) = \{ T(\vec{x}) : \vec{x} \in \mathbb{R}^n \}$$

2. Matrix Transformations

Def: Given a matrix A we can define a **linear transformation** $T: \mathbb{R}^n \to \mathbb{R}^m$ as follows:

($A \in \mathbb{R}^{m \times n}$, $T: \mathbb{R}^n \to \mathbb{R}^m$, $\vec{x} \in \mathbb{R}^n$)

$$T(\vec{x}) = A\vec{x}$$

Thm: $A \in \mathbb{R}^{m \times n}$, T is the matrix transformation $T(\vec{x}) = A\vec{x}$ then the range of T is equal to the span of columns of A

3. Linear Transformation

Def: A transformation T is called a linear transformation if

(1) $T(\vec{x}+\vec{y}) = T(\vec{x}) + T(\vec{y})$ for all $\vec{x}, \vec{y}$ in the domain of T

(2) $T(c\vec{x}) = cT(\vec{x})$ for all scalars $c \in \mathbb{R}$ and $\vec{x}$ in the domain of T

The: If T is any linear transformation then,

$$T(c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_n\vec{v}_n) = c_1T(\vec{v}_1) + c_2T(\vec{v}_2) + \cdots + c_nT(\vec{v}_n)$$

for any vectors $\vec{v}_1, \ldots, \vec{v}_n$ and scalars $c_1, \ldots, c_n$

In particular, $T(\vec{0}) = \vec{0}$

The: Every matrix transformation is linear transformation

① addition:

$$A(\vec{u}+\vec{v}) = A\vec{u} + A\vec{v}$$

$$\Rightarrow \quad T(\vec{u}+\vec{v}) = T(\vec{u}) + T(\vec{v})$$

② scale multiplication:

$$c(A\vec{u}) = A(c\vec{u})$$

$$\Rightarrow cT(\vec{u}) = T(c\vec{u})$$